SPOT-The-Difference: An Independent Evaluation of Latent-Aware Causal Structure Learning for Microservice Latency on PetShop

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Abstract*—Milestone 3 evaluates Skeleton Posterior-guided Optimization (SPOT) against its ABIC backbone on simulated ancestral acyclic directed mixed graphs (ADMGs) and the PETSHOP microservice benchmark. The evaluation extends the partial M2 study with four PetShop scenarios, paired optimizer seeds, block-bootstrap resampling, ablations, classical FCI/RFCI baselines, posterior-quality measurements, directed-only experiments, preprocessing sensitivity, and root-cause analysis (RCA). On synthetic graphs, calibrated SPOT improves skeleton F1 over ABIC by 0.045–0.088 across sample sizes; the gain is statistically supported at $n = 200, 1000, 2000$, but not at $n = 500$. On PETSHOP, pooled calibrated-SPOT skeleton F1 is 0.542 versus 0.513 for ABIC and PAG structural Hamming distance (SHD) is 44.95 versus 48.85. The four scenario means favour SPOT, but the high-traffic paired difference is only 0.003 and five optimizer seeds do not establish a reliable per-scenario accuracy effect. The strongest PetShop evidence comes from the block bootstrap: high-traffic skeleton F1 improves by 0.038 (17 wins in 20 paired resamples, $p = 0.003$), while the other two tested scenarios are not distinguishable from zero. Ablations indicate that the posterior contains useful information: a random posterior collapses skeleton F1 to 0.196, whereas an oracle posterior reaches 0.813. FCI/RFCI outperform both methods on the synthetic benchmark and are competitive on PetShop. The learned posterior is strong on simulated graphs (AUROC 0.990, log-loss 0.052) but weaker on PetShop (AUROC 0.84–0.92); post-hoc calibration does not improve it. RCA does not show that learned graphs outperform a no-graph anomaly baseline. Experiments E6 (equal wall-clock budget) and E8 (rolling-split inference validation) were not run and are explicitly left unevaluated. Overall, the evidence supports a limited, data-dependent benefit rather than a general claim that SPOT is superior.

***Index Terms*—causal discovery, latent confounding, ADMG, PAG, skeleton posterior, SPOT, ABIC, PetShop, microservices, root-cause analysis**

I. INTRODUCTION

The M2 project established a pipeline for learning latent-aware causal structure from microservice latency measurements. The benchmark is PETSHOP, which supplies service-level metrics, a call graph, and injected-issue runs. Some services have no usable metric stream, making latent variables a property of the application rather than an artificial deletion. The project therefore represents observed services with an ADMG: directed edges represent measured causal propagation, while bidirected edges represent dependence induced by unobserved common causes.

M3 changes the emphasis from demonstrating that SPOT can outperform ABIC to identifying when posterior-guided optimization helps, when it does not, and whether the learned structure is useful for downstream RCA. This framing follows the course requirement to evaluate the method honestly rather than force a positive result.

The M3 evaluation addresses five hypotheses fixed in the project plan:

- H1: On simulated ancestral ADMGs, SPOT improves skeleton accuracy over ABIC, with the gain expected to shrink as sample size grows.
- H2: On PETSHOP, the gain is smaller than on synthetic data because the skeleton posterior is out of distribution.
- H3: The speed-up of SPOT is driven mainly by the hard pruning floor rather than by the stochastic mask.
- H4: The skeleton posterior is well calibrated on simulated data and over-confident on PETSHOP.
- H5: A fitted ADMG predicts held-out data better than a no-structure baseline, and accurate structure helps RCA.

A difference is treated as supported only when a paired test has $p < 0.05$ and its 95% confidence interval excludes zero. When this condition is not met, the result is described as inconclusive or not distinguishable from noise.

II. DATASET AND GROUND-TRUTH CONSTRUCTION

A. PetShop

We retain the four scenarios: `low_traffic`, `high_traffic`, `temporal_traffic1`, and `temporal_traffic2`. As in M2, latency averages from normal-operation (`noissue`) runs are used for structure learning; injected-issue runs are reserved for RCA. The causal direction reverses the application call direction because callee latency propagates to callers.

A service is observed when it has a latency column with at most 5% missing values and non-constant values. The remaining services are treated as latent. Ground truth is obtained by (1) reversing the call graph, (2) applying Verma–Pearl latent projection onto observed services, (3) obtaining the maximal

ancestral graph (MAG), and (4) converting the MAG to a partial ancestral graph (PAG) for scoring. The resulting scenarios contain approximately 29–35 observed variables and 9–13 latent services. Their true PAGs contain 30–38 directed edges and zero or one bidirected edge.

The benchmark contains one dataset per scenario. Consequently, repeated seeds measure optimizer-start variability, not independent draws of new PetShop datasets. This distinction is important when interpreting confidence intervals and paired tests.

B. Preprocessing

The primary PetShop experiments use Gaussian-rank preprocessing. M2 documented that latency is right-skewed and that Gaussian-rank transformation makes the marginal distributions more compatible with the linear-Gaussian model, while potentially distorting scale relationships. Time-series observations are treated as independent by the learner despite measurable lag-1 autocorrelation. This is a limitation shared by the compared methods. E13 separately evaluates `raw_centered` preprocessing; its results are not used to replace the primary Gaussian-rank results.

III. Model and Learning Methods

A. ADMG representation

Both methods fit a linear-Gaussian structural equation model with correlated errors:

$$X = D^\top X + \epsilon, \qquad \epsilon \sim \mathcal{N}(0, \Omega), \tag{1}$$

with covariance

$$\Sigma = (I - D^\top)^{-1} \Omega (I - D^\top)^{-\top}. \tag{2}$$

The support of D represents directed edges and the off-diagonal support of Ω represents bidirected edges. The learned ADMG is converted through a MAG to a PAG before comparison with the ground-truth PAG.

B. ABIC and SPOT

ABIC is the unguided baseline and the optimization backbone. It minimizes a least-squares data term with a smooth-ℓ_0 penalty under acyclicity and ancestrality constraints. The constraints are handled with an augmented Lagrangian and L-BFGS-B, with RICF updates for the error-covariance pseudo-variables.

SPOT uses a skeleton posterior $p_{ij} = q_\phi(S_{ij} = 1 \mid X)$, trained offline on simulated ancestral ADMGs. During optimization, the gradient for a pair is masked according to the posterior and the direction of the proposed update. A hard floor can additionally pin low-posterior pairs to zero. We report the calibrated floor 6×10^{-3} and the released-code floor 10^{-3}. The methods share the optimizer, objective, restarts, thresholds, and seeds; the posterior-guided mask and optional floor are the intended differences.

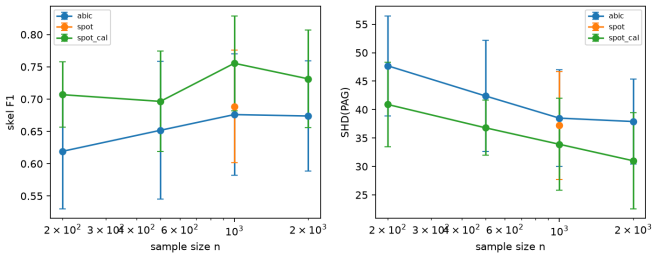


Fig. 1. E1 synthetic sample-size sweep: skeleton F1 for ABIC and calibrated SPOT.

IV. Experimental Protocol

The experiments compare ABIC, calibrated SPOT, and, where applicable, released-floor SPOT. Synthetic experiments use simulated ancestral ADMGs. PetShop experiments use all four scenarios and paired optimizer seeds. E3 uses five seeds per scenario. E4 ablations use three seeds per scenario. E5 uses 20 paired moving-block bootstrap resamples for each of three scenarios. FCI and RFCI are evaluated at significance levels $\alpha \in \{0.01, 0.05\}$.

The principal KPIs are skeleton, arrowhead, and tail F1; PAG SHD; estimated directed and bidirected edge counts; and optimization time. Posterior quality is evaluated using AUROC, AUPRC, Brier score, log-loss, and expected calibration error (ECE). RCA uses top-1 recall, top-3 recall, and mean reciprocal rank (MRR).

Timing is reported cautiously. Runs were executed on a contended machine and absolute times varied. Runtime ratios are more informative than absolute seconds, but they are still affected by system load. In particular, the pooled E3 optimization-time ratio should be interpreted as an indicative comparison rather than a hardware-independent speed claim.

V. Results

A. E1: Synthetic sample-size sweep

For $d = 25$, ten random graphs were evaluated at each sample size $n \in \{200, 500, 1000, 2000\}$. Calibrated SPOT improves skeleton F1 at every sample size. The mean gains over ABIC are approximately $+0.088, +0.044, +0.080, +0.057$, respectively. Paired tests support gains at $n = 200, 1000, 2000$, but not at $n = 500$. The effect is positive throughout, but it does not decrease monotonically with sample size; therefore, the second part of H1 is not cleanly established.

B. E2: Synthetic robustness

At $n = 1000$, E2 varies graph density, measurement noise, autocorrelation, and missingness. The largest reported gain occurs in the sparse setting (density 1), where calibrated SPOT improves skeleton F1 by approximately 0.093. At high signal-to-noise ratio, the gain is small (approximately 0.007). Across the tested settings the effect is therefore conditional on the data regime, not a uniform improvement. The missing-data

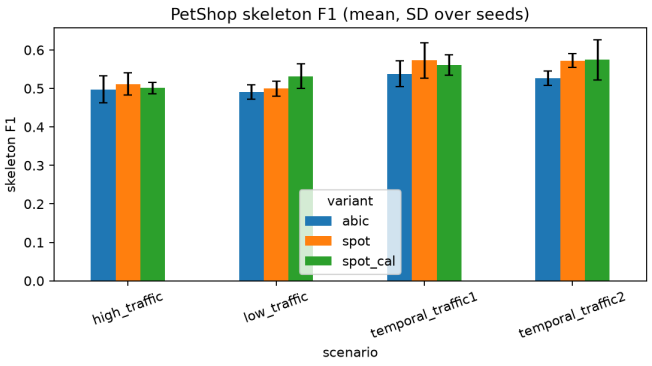


Fig. 2. E3 PetShop skeleton F1 across the four scenarios.

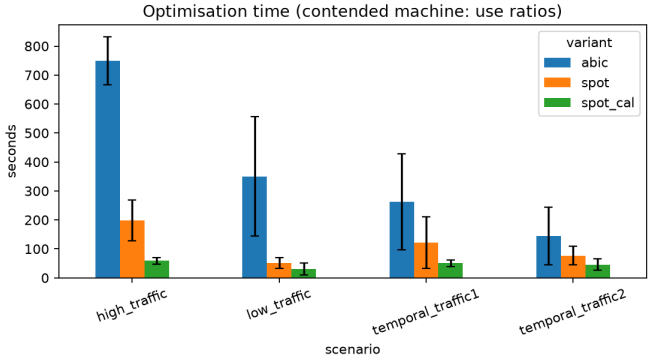


Fig. 3. E3 optimization time by scenario. Timing ratios are indicative because machine load varied.

experiments include 5% and 20% MCAR settings with mean imputation; autocorrelation settings include 0, 0.3, and 0.6.

C. E3: PetShop main comparison

The detailed scenario-wise results are reported in Table I. Across four scenario means, calibrated SPOT reaches skeleton F1 0.542 compared with 0.513 for ABIC (difference $+0.029$). PAG SHD improves from 48.85 to 44.95. Skeleton F1 favours calibrated SPOT in all four scenario means. However, high traffic is effectively a tie: the paired difference is only $+0.003$, with two wins and three losses over five seeds. With five paired seeds, the smallest attainable two-sided exact Wilcoxon p-value is approximately 0.062; per-scenario evidence is consequently limited.

The pooled optimization times are approximately 376 s for ABIC and 46 s for calibrated SPOT, an indicative ratio of about 8.1. These timings were measured on a contended machine and vary substantially across scenarios; they should not be interpreted as a stable universal speed-up. The temporal scenarios show smaller runtime differences than low- and high-traffic runs.

D. E5: Block-bootstrap comparison

The paired bootstrap results are summarized in Table II. E5 provides the strongest PetShop evidence because it compares

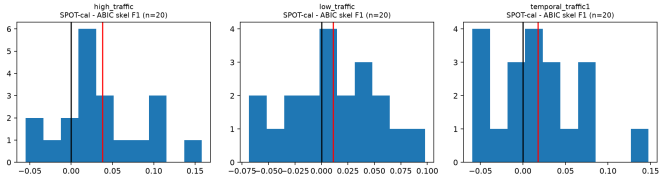


Fig. 4. E5 paired bootstrap distribution of skeleton-F1 differences.

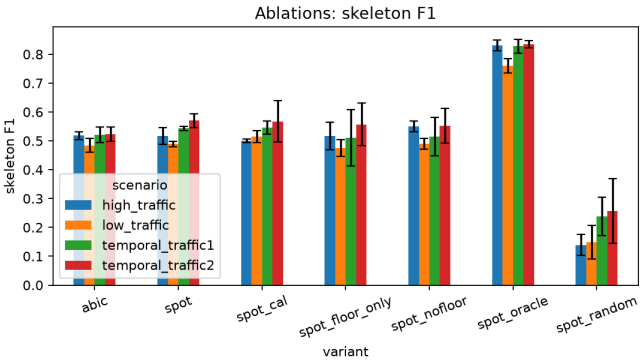


Fig. 5. E4 PetShop ablation results.

paired methods on the same moving-block resamples. In high traffic, calibrated SPOT improves skeleton F1 by $+0.038$, wins 17 of 20 resamples, and loses 3; the reported paired test gives $p = 0.003$. PAG SHD decreases by 4.8. This is evidence of a gain for this scenario and resampling design. In low traffic, the gain is $+0.011$, and in temporal traffic 1 it is $+0.018$; neither is distinguishable from zero under the predeclared decision rule. Bootstrap resamples are dependent blocks from a single scenario dataset, so they do not create 20 independent PetShop deployments.

E. E4: Ablations

The PetShop ablations separate the stochastic mask, hard floor, random posterior, and oracle posterior. The ordinary variants have skeleton F1 between 0.515 and 0.532, with standard deviations of roughly 0.04–0.07; their accuracy differences are not reliably separable. The mask-only variant takes about 140 s compared with about 394 s for ABIC (approximately 2.8 times faster), while adding the calibrated floor reduces the time to about 49 s. This pattern supports the interpretation that pruning contributes materially to the speed-up, although the ablation does not establish a clean causal decomposition of all runtime effects.

The random-posterior control collapses skeleton F1 to 0.196, showing that arbitrary masking is not sufficient. The oracle posterior reaches 0.813, demonstrating substantial headroom when the skeleton information is accurate. The gap between the learned and oracle posterior therefore points to posterior quality as a key limitation, rather than an optimizer ceiling.

TABLE I

PETSHOP MAIN COMPARISON (E3): MEAN ± SD OVER FIVE OPTIMIZER SEEDS. GAUSSIAN-RANK DATA, ONE RESTART, AND NO WALL-CLOCK BUDGET. OPTIMIZATION TIMES WERE MEASURED ON A SHARED MACHINE AND SHOULD BE READ AS INDICATIVE RATIOS RATHER THAN STABLE ABSOLUTE TIMINGS.

Scenario	Method	Skeleton F1	Arrowhead F1	Tail F1	SHD(PAG)	Opt. time (s)
low_traffic	ABIC	0.491 ± 0.018	0.265 ± 0.032	0.025 ± 0.056	43.8 ± 2.5	350 ± 205
	SPOT (10^{-3})	0.500 ± 0.019	0.272 ± 0.030	0.000 ± 0.000	41.8 ± 1.6	52 ± 19
	SPOT-cal (6×10^{-3})	0.532 ± 0.031	0.265 ± 0.049	0.000 ± 0.000	39.4 ± 2.1	30 ± 20
high_traffic	ABIC	0.498 ± 0.035	0.258 ± 0.048	0.077 ± 0.070	61.2 ± 4.1	749 ± 83
	SPOT (10^{-3})	0.512 ± 0.029	0.268 ± 0.014	0.162 ± 0.123	57.8 ± 0.8	198 ± 70
	SPOT-cal (6×10^{-3})	0.501 ± 0.015	0.264 ± 0.036	0.149 ± 0.103	56.6 ± 2.3	59 ± 12
temporal_traffic1	ABIC	0.538 ± 0.033	0.354 ± 0.049	0.031 ± 0.069	44.2 ± 2.8	262 ± 166
	SPOT (10^{-3})	0.573 ± 0.046	0.356 ± 0.033	0.078 ± 0.072	41.6 ± 3.6	122 ± 89
	SPOT-cal (6×10^{-3})	0.561 ± 0.026	0.374 ± 0.050	0.053 ± 0.073	42.0 ± 2.3	50 ± 11
temporal_traffic2	ABIC	0.526 ± 0.019	0.295 ± 0.047	0.027 ± 0.060	46.2 ± 5.2	144 ± 99
	SPOT (10^{-3})	0.572 ± 0.018	0.359 ± 0.021	0.054 ± 0.074	43.0 ± 1.6	77 ± 31
	SPOT-cal (6×10^{-3})	0.574 ± 0.052	0.366 ± 0.076	0.080 ± 0.074	41.8 ± 4.4	46 ± 19
Mean of scenario means	ABIC	0.513	0.293	0.040	48.9	376
	SPOT (10^{-3})	0.539	0.314	0.073	46.1	112
	SPOT-cal (6×10^{-3})	0.542	0.317	0.071	45.0	46

TABLE II

E5 MOVING-BLOCK BOOTSTRAP: PAIRED SPOT-CAL MINUS ABIC DIFFERENCES OVER 20 RESAMPLES.

Scenario	Metric	ABIC	SPOT-cal	Difference [95% CI]	p_W
low	Skeleton F1	.488	.499	$+.011[-.010, .032]$	.314
low	SHD(PAG)	44.2	43.0	$-1.25[-2.9, .4]$	.159
high	Skeleton F1	.466	.505	$+.038[.014, .062]$	.003
high	SHD(PAG)	65.0	60.2	$-4.8[-6.9, -2.8]$	.001
temporal1	Skeleton F1	.518	.535	$+.018[-.006, .042]$	.171
temporal1	SHD(PAG)	44.6	42.3	$-2.35[-4.9, .2]$	.101

F. E7: Posterior probability quality

On held-out simulated graphs, the posterior reaches AUROC 0.990 and log-loss 0.052. On PetShop, AUROC ranges approximately from 0.84 to 0.92 and log-loss from 0.26 to 0.32. There is only one graph per PetShop scenario, so these scenario-level posterior metrics do not provide replication across independent graphs. Isotonic and Platt calibration do not improve the overall result; isotonic calibration is worse on log-loss in the PetShop scenarios. The results support strong discrimination on simulated data but weaker transfer to PetShop. They do not establish that the posterior is uniformly over-confident in every calibration metric; the more defensible conclusion is distribution shift and degraded probability quality on PetShop.

G. E9: FCI and RFCI baselines

E9 materially changes the project's comparative positioning. On synthetic data, FCI/RFCI outperform SPOT on both skeleton and arrowhead F1. Across sample sizes, their skeleton F1 is approximately 0.79–0.91, compared with approximately 0.71–0.76 for SPOT in the corresponding comparison. Arrowhead F1 is approximately 0.48–0.74 for FCI/RFCI, compared with approximately 0.37–0.46 for SPOT. Thus, SPOT's advantage over ABIC does not imply superiority over classical constraint-based methods.

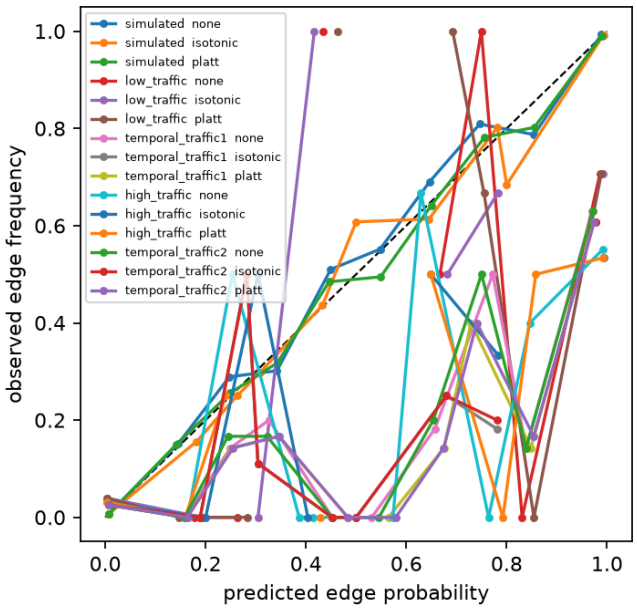


Fig. 6. E7 reliability diagrams for simulated and PetShop data.

On PetShop, FCI is broadly comparable to calibrated SPOT: skeleton F1 is approximately 0.45–0.56 for FCI and 0.50–0.57 for calibrated SPOT across scenarios. FCI is better on

tail marks in the temporal scenarios. These results make the classical baselines essential to the final evaluation rather than optional context.

H. E10: Directed-only learner

Forcing the bidirected block to zero reduces arrowhead F1 from approximately 0.3 in the full model to approximately 0.05 in the directed-only comparison, while skeleton F1 does not show a clear consistent change. This indicates that the model's orientation performance is closely tied to its treatment of bidirected edges. It does not mean that the learned bidirected structure is correct: the full learners substantially overproduce bidirected edges relative to the ground truth.

I. E12: Root-cause analysis

The RCA evaluation includes 36 cases and three graph seeds. These cases are not independent in the sense required to treat all rows as independent experimental units. The true graph obtains top-3 recall of approximately 0.83–0.86, while the no-graph anomaly score reaches approximately 0.81. Learned graphs do not beat this no-graph baseline: the ABIC DAG is approximately 0.82 at top-3, while calibrated-SPOT DAG is approximately 0.67. Restricting candidates to ancestors only reduces top-3 performance to approximately 0.41–0.53. The result does not support H5's claim that learned structure improves RCA over a no-structure baseline. It also cautions against assuming that a structurally motivated ancestor filter necessarily helps operational diagnosis.

J. E13: Preprocessing sensitivity

Under `raw_centered` preprocessing, both methods perform substantially worse, with skeleton F1 approximately 0.15–0.44 across the reported scenario-method combinations. The relative SPOT advantage is larger in this setting, but the low absolute scores show that conclusions are specific to the Gaussian-rank preprocessing used in the main PetShop experiments. E13 is a sensitivity result, not a replacement for the primary evaluation.

VI. Experiments Not Run

Two planned experiments were not completed:

- **E6 – equal wall-clock budget:** the planned 60 s and 120 s comparisons were not run. No equal-budget accuracy claim is made in this report.
- **E8 – rolling-split inference validation:** held-out NLL, RMSE, 90% interval coverage, and inference-error-versus-time measurements on rolling splits were not run. The report therefore does not claim validated predictive inference performance from E8.

The absence of these experiments limits conclusions about fair equal-time accuracy and held-out probabilistic prediction. They are not silently inferred from other experiments.

TABLE III
M3 hypothesis assessment

H	Assessment	Evidence and qualification
H1	Partly supported	Positive synthetic skeleton-F1 gains at all n; statistically supported at $n = 200, 1000, 2000$, not 500. No monotonic shrinkage with n.
H2	Supported	PetShop gain is modest; posterior AUROC falls from 0.990 simulated to 0.84–0.92 on PetShop. Scenario-level evidence is limited.
H3	Partly supported	Mask-only reduces time; the floor reduces it further. Runtime attribution is not fully isolated.
H4	Partly supported	Strong simulated AUROC/log-loss; weaker PetShop performance. Calibration methods do not improve the PetShop results.
H5	Not supported	Learned graphs do not outperform the no-graph RCA baseline; ancestor-only filtering hurts. E8 was not run.

VII. Discussion and Revised SOTA Positioning

The results support a narrower claim than the original project ambition. On simulated data, calibrated SPOT improves skeleton F1 over its ABIC backbone, although the gain is not significant at every sample size and does not monotonically shrink as n increases. On PetShop, the pooled skeleton-F1 gain is modest. The bootstrap identifies a clear high-traffic improvement, but low traffic and temporal traffic 1 remain inconclusive. This is evidence of a scenario-dependent benefit, not a general win.

The ablations show that the posterior is informative: randomizing it is damaging and an oracle posterior performs substantially better. However, the learned posterior transfers imperfectly from simulated graphs to PetShop. The hard floor contributes substantially to runtime reduction, and the stochastic mask alone already provides a speed-up. These observations refine H3: pruning is an important contributor, but the observed timing differences are not a clean isolated causal estimate because optimizer behavior and machine contention also matter.

The FCI/RFCI results are a significant counterweight. They outperform SPOT on synthetic data and remain competitive on PetShop. Therefore, the project cannot claim that SPOT is state of the art or that it is the best general-purpose latent causal discovery method. A defensible contribution is an independent, corrected, application-specific evaluation of posterior-guided optimization on a real microservice benchmark, including an analysis of its transfer limits and comparison against classical baselines.

A. Hypothesis scorecard

VIII. Limitations

First, each PetShop scenario is one dataset. Optimizer seeds and bootstrap blocks quantify variability conditional on that dataset; they do not represent independent benchmark instances. Second, the learner assumes an i.i.d. linear-Gaussian

model, while latency series are autocorrelated and may be non-Gaussian. Third, the ground truth is derived from the inverted call graph and latent projection; it expresses the designed propagation structure and is not guaranteed to equal every statistical dependency in observed measurements. Fourth, both methods overuse bidirected edges relative to the truth, complicating arrowhead and tail interpretation. Fifth, runs use limited restarts and timings were affected by machine contention. Finally, E6 and E8 were not run, and no result should be inferred for them.

IX. Conclusion

M3 finds that SPOT offers a real but conditional improvement over ABIC. Synthetic results favour SPOT, and the PetShop block bootstrap gives strong evidence of a high-traffic skeleton-F1 gain. The pooled PetShop improvement is modest, and the high-traffic five-seed comparison is effectively tied. Ablations show that posterior information matters and that the oracle posterior leaves substantial headroom, while the posterior's transfer to PetShop remains imperfect. The hard floor contributes to the observed speed-up. Classical FCI/RFCI outperform SPOT on synthetic graphs and are competitive on PetShop, while the RCA results do not show a benefit over the no-graph anomaly baseline. The appropriate conclusion is therefore not that SPOT wins, but that posterior-guided optimization can help in selected regimes and that posterior transfer, baseline selection, and downstream utility remain open challenges.

Acknowledgment

The project code and report preparation used generative-AI assistance. All team members are responsible for understanding and verifying the experiments, metrics, statistical tests, and conclusions reported here.

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